Experiment ‘p_3_6_parametrised_order’ Results

June 3, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` **:** 0
Responses containing `expect` **:** 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_3_6_parametrised_order
Natural language statement: Write a method that takes three numbers and returns ”increasing” if they are in increasing order, ”decreasing” if they are in decreasing order, and ”neither” otherwise. A Boolean parameter triggers the method to be ”strict” or ”lenient”. In lenient mode, the sequence 3 4 4 is increasing and the sequence 4 4 4 is both increasing and decreasing.
Method signature: p_3_6_parametrised_order (i: int, j: int, k: int, strict: bool) returns (s: string)

Ensures

- `if strict then if i < j < k then s == "increasing" else if i > j && j > k then s == "decreasing" else s == "neither" else if i <= j <= k then if i == j == k then s == "increasing and decreasing" else s == "increasing" else if i >= j && j >= k then s == "decreasing" else s == "neither"`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method that takes three numbers and returns "increasing" if they are in
    ↪ increasing order, "decreasing" if they are in decreasing order, and "neither"
    ↪ otherwise. A Boolean parameter triggers the method to be "strict" or "lenient". In
    ↪  lenient mode, the sequence 3 4 4 is increasing and the sequence 4 4 4 is both
    ↪ increasing and decreasing.

The signature should be:

p_3_6_parametrised_order (i: int, j: int, k: int, strict: bool ) returns (s: string)

The method should respect the following contract:

ensures if strict then if i < j < k then s == "increasing" else if i > j && j > k then s
    ↪ == "decreasing" else s == "neither" else if i <= j <= k then if i == j == k then s
    ↪  == "increasing and decreasing" else s == "increasing" else if i >= j && j >= k
    ↪ then s == "decreasing" else s == "neither"

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DmdqJ8ckgwmaHY66FcRedH7Yd9sJE
Model: gpt-5.5-2026-04-23
Created at: 1780485271
Finish reason: stop
Usage: 596 tokens in, and 693 tokens out

```
{
  if strict {
    if i < j && j < k {
      s := "increasing";
    } else if i > j && j > k {
      s := "decreasing";
    } else {
      s := "neither";
    }
  } else {
    if i <= j && j <= k {
      if i == j && j == k {
        s := "increasing and decreasing";
      } else {
```

```
        s := "increasing";
      }
    } else if i >= j && j >= k {
      s := "decreasing";
    } else {
      s := "neither";
    }
  }
}
```

Final Program

```
method p_3_6_parametrised_order (i: int, j: int, k: int, strict: bool ) returns (s:
    ↪ string)
        ensures if strict then if i < j < k then s == "increasing" else if i > j && j > k
            ↪ then s == "decreasing" else s == "neither" else if i <= j <= k then if i ==
            ↪  j == k then s == "increasing and decreasing" else s == "increasing" else
            ↪ if i >= j && j >= k then s == "decreasing" else s == "neither"
{
  if strict {
    if i < j && j < k {
      s := "increasing";
    } else if i > j && j > k {
      s := "decreasing";
    } else {
      s := "neither";
    }
  } else {
    if i <= j && j <= k {
      if i == j && j == k {
        s := "increasing and decreasing";
      } else {
        s := "increasing";
      }
    } else if i >= j && j >= k {
      s := "decreasing";
    } else {
      s := "neither";
    }
  }
}
```

Total Token Usage

Input tokens: 596
Output tokens: 693
Reasoning tokens: 512
Sum of 'total tokens': 1289

Experiment Timings

Overall Experiment started at 1780485271283, ended at 1780485287247, lasting 15964ms (15.96 seconds)
Iteration #1 started at 1780485271284, ended at 1780485287247, lasting 15963ms (15.96 seconds)